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Testing VPC Connectivity

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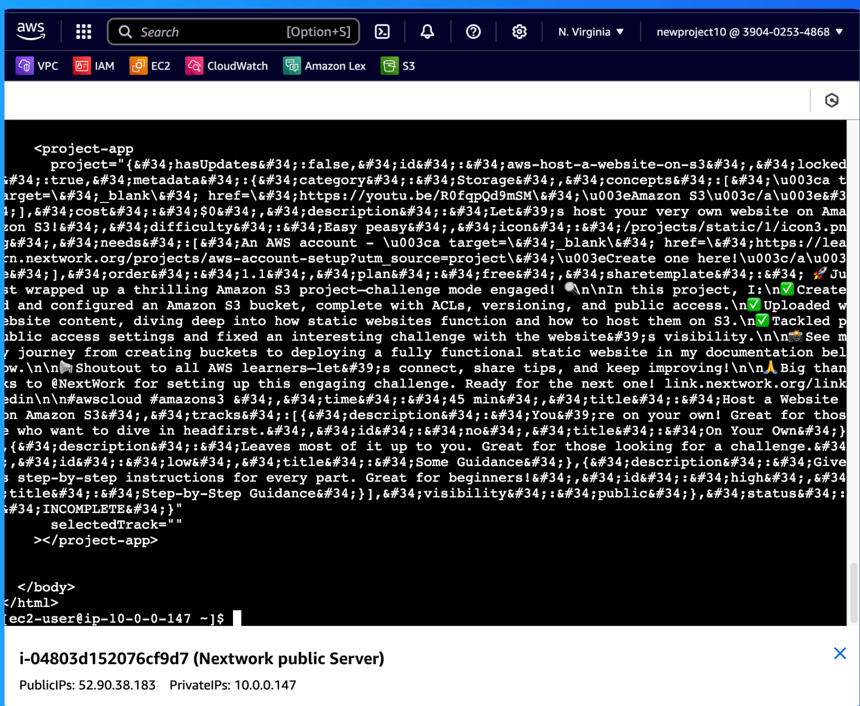

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Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud (VPC) is a virtual network environment within the AWS cloud that you can use to launch AWS resources like EC2 instances, databases, and other services. It's essentially a private, isolated network that you can customize.

How I used Amazon VPC in this project

I was able to utilize my Amazon VPC in todays project to test connectivity between private and public servers. It allowed to to showcase my abilities in troubleshooting connectivity issue by editing security groups, ACL's and subnets.

One thing I didn't expect in this project was...

I did not expect to see how possible it is to allow and block the diffrent type of traffic that is allowed to reach a VPC.

This project took me...

I spend a total of 1 hour on this project.

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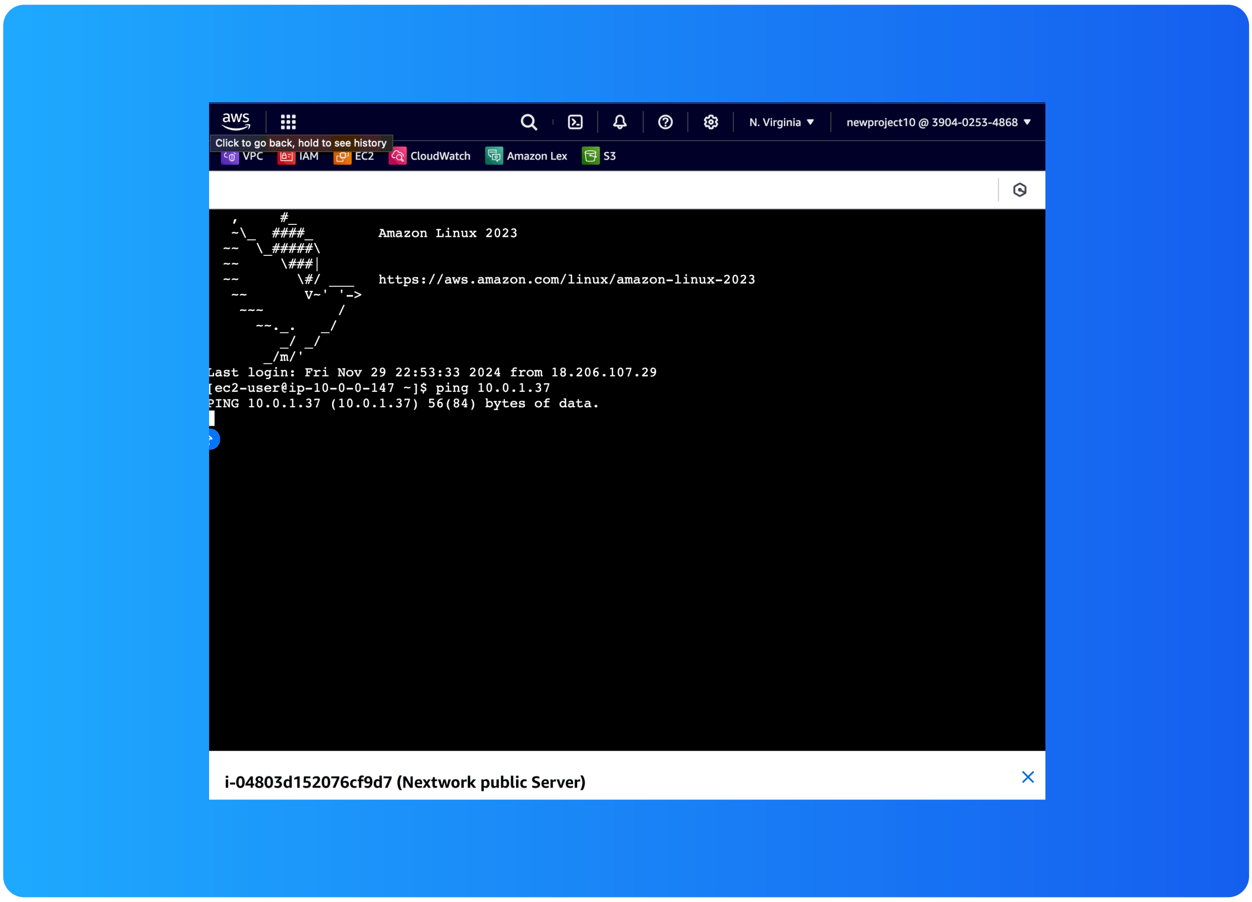
Connecting to an EC2 Instance

Connectivity means getting resources in our network to communicate with with each other, and how well they can communicate delivered data to each other. Without connectivity resources in our network would not be able to communicate.

My first connectivity test was whether I could connect to my networks public EC2 instance.



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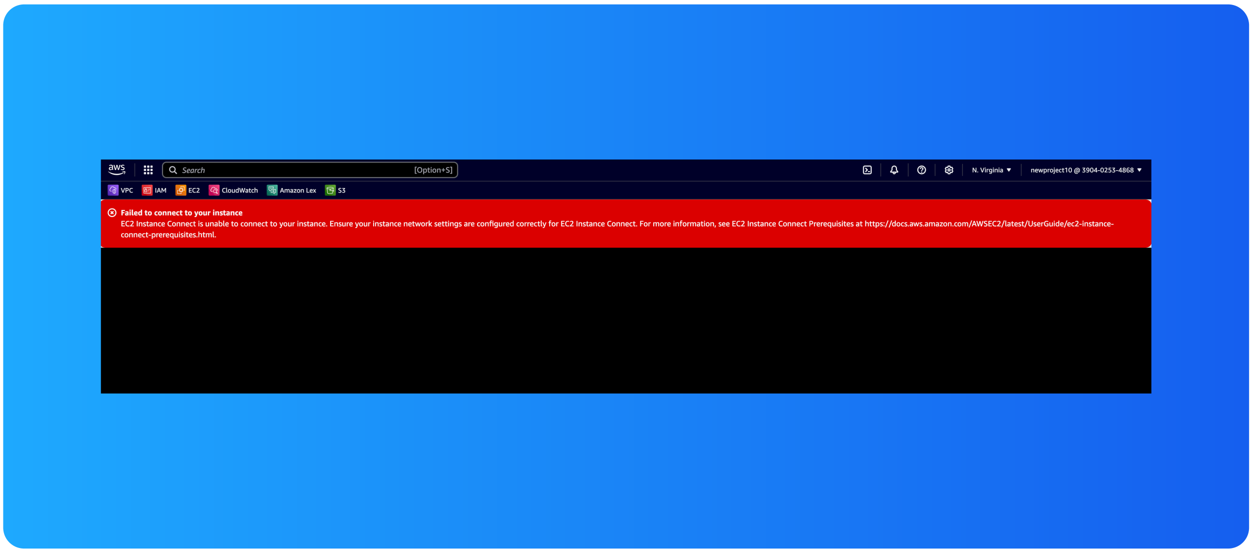
EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is when one EC2 Instance makes a connection with another EC2 Instance without any errors. This allow for communication to happen between the two in a private way.

My first attempt at getting direct access to my public server resulted in an error, because my private server has a security group that did not allow SSH traffic, it only allow HTTP traffic a different protocol.

I fixed this error by adding a new inbound rule in my private servers security group that allowed SSH traffic from anywhere.



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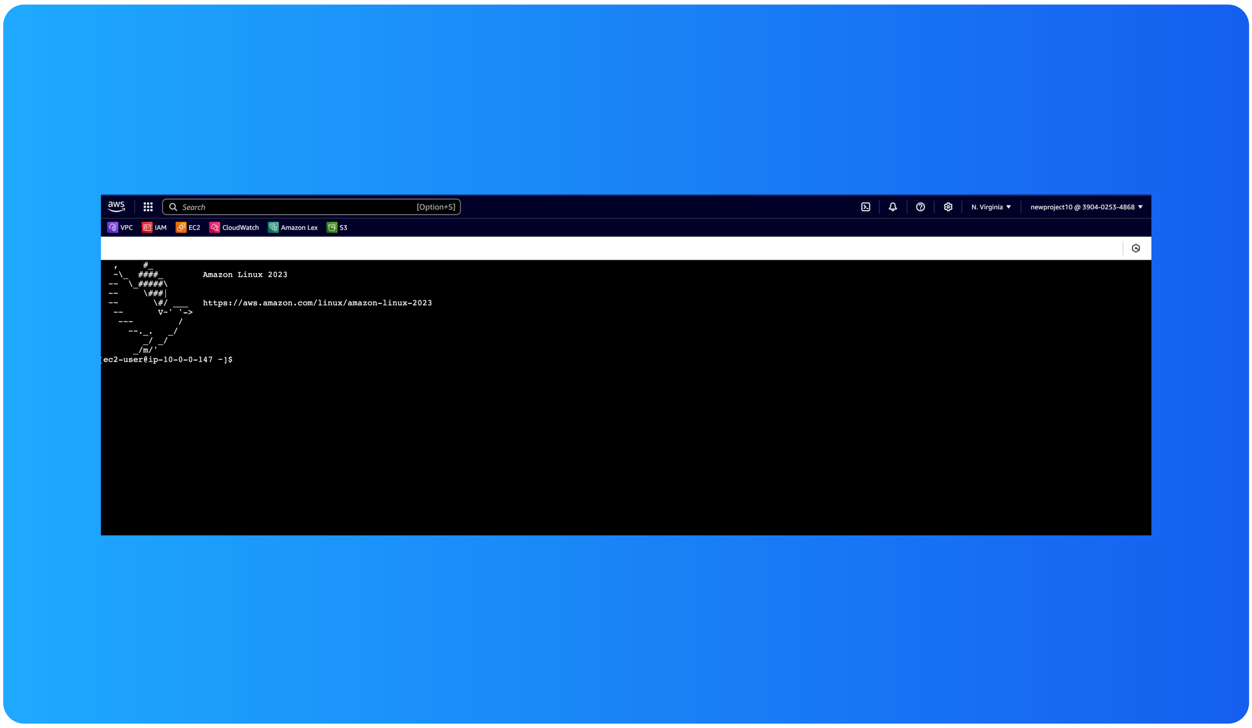
Connectivity Between Servers

Ping is a tool i used to test connectivity between two servers and also the response times. I used ping to test the connectivity between my public and private servers.

The ping command I ran was ping followed by the private ipv4 address for our private server which was 10.0.0.147

The first ping returned no replies from the private server. This meant security settings with my private server was blocking our inbound and outbound icmp traffic. Which is the traffic type of ping messages.



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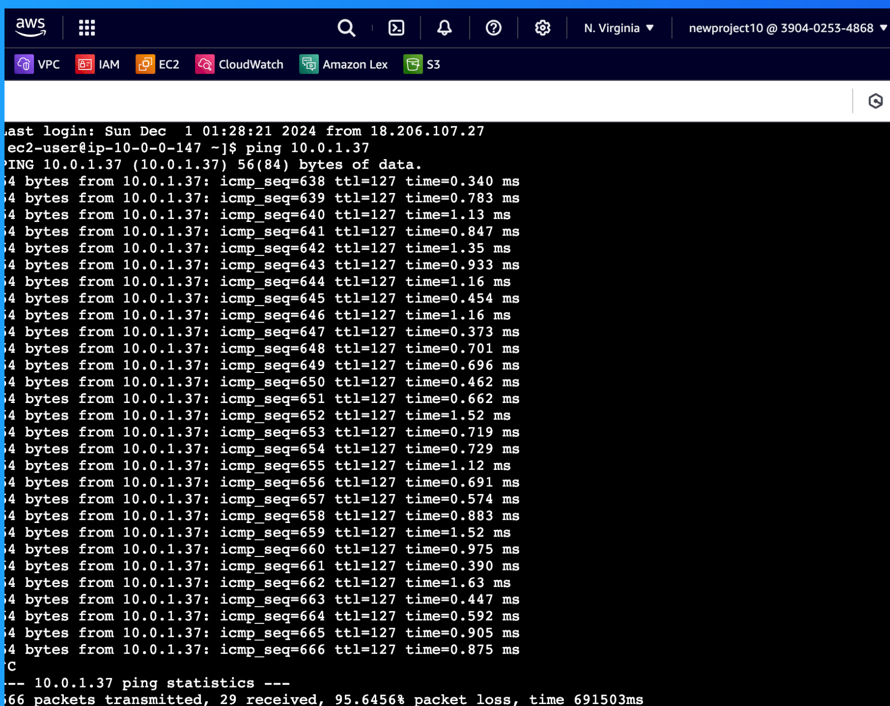
Troubleshooting Connectivity

I troubleshooted this by allowing icmp traffic to flow in my private servers network ACL's and security groups.



Connectivity to the Internet

Curl is a command we use to see if we can communicate with internet. The same we use ping to communicate with another server curl allow us to see if there is a responce from a internet host.

I used curl to test if the connectivity of my public server, security group and network ACL was set up properly. Access to the webpage showed there is communication with the internet.

Ping vs Curl

Ping allows test network connectivity and measure latency (round-trip time) between two devices. While curl is primarily used to fetch data from web servers, test APIs, and simulate HTTP requests.

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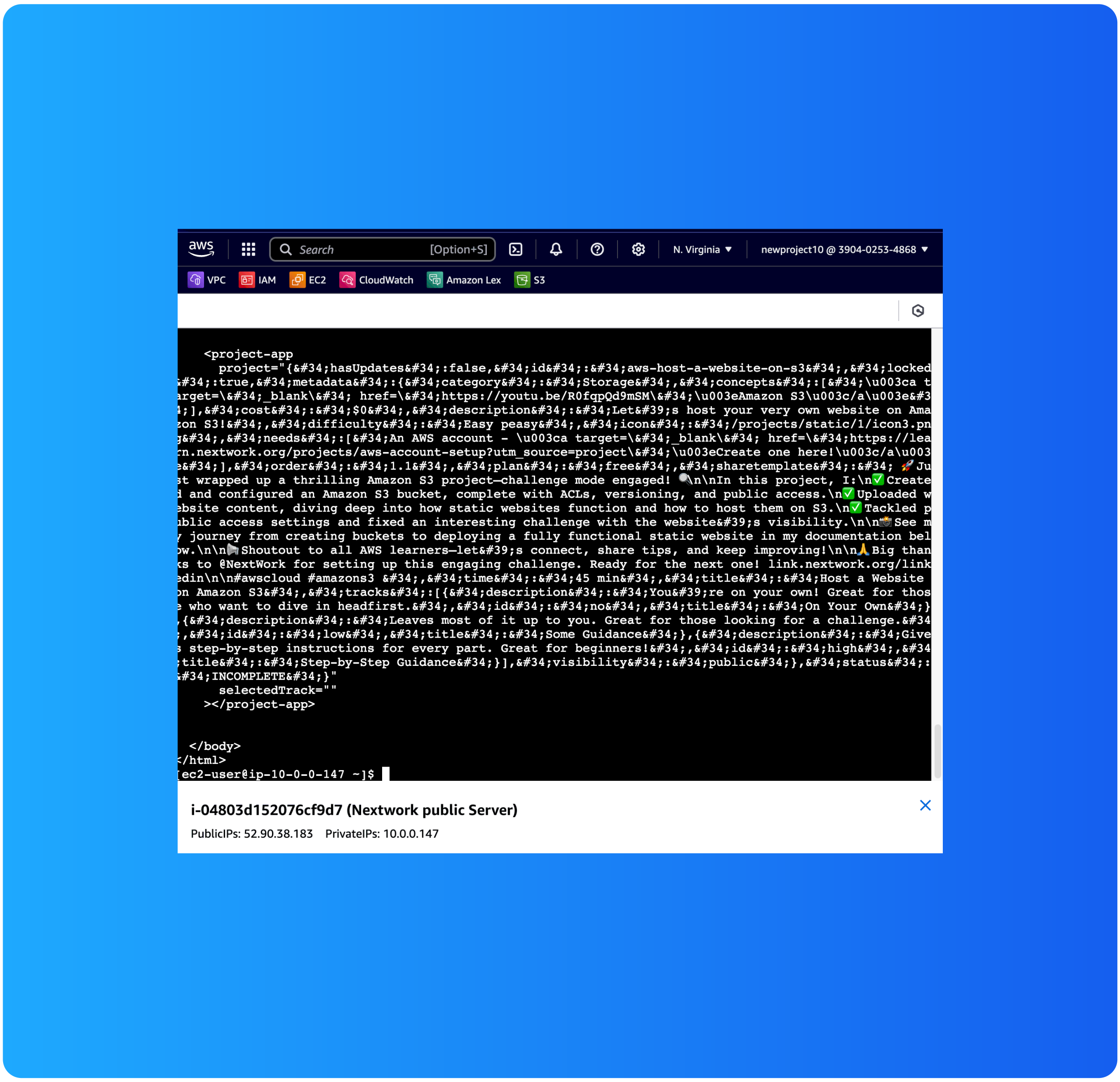
Connectivity to the Internet

The curl command I used was "curl https://learn.nextwork.org/projects/aws-host-a-website-on-s3" Primarily used to fetch data from web servers, test APIs, and simulate HTTP requests.



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